

$\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ | Amorphous Oxychloride

Matched-temperature comparison of pretrained-potential MD with experiment and AIMD

1 Literature

Hu et al. (2023): experiment
Hussain et al. (2024): AIMD
Reference T: 340 / 360 / 380 K

2 Model & software

Integer-occupancy glass
192 atoms; periodic cell
Pretrained NEP89
GPUMD

3 MD conditions

T: 340 / 360 / 380 K
Ensemble: fixed-cell NVT
Thermostat: NHC; $\tau_T = 100$ fs
dt: 2 fs
Production: 300 ps

4 Analysis

Li / host MSD $\rightarrow$ D(T)
RDF and coordination
Radial displacement
Direct AIMD D comparison

5 Conclusion

NEP89 D below AIMD
Local geometry persists
Mechanism not yet unique